

Yes. For **PS-33**, I would make the demo much more than an AI image classifier. The judge should experience a complete journey:

 **Identify** →  **Understand** →  **Value** →  **Impact** →  **Pickup** → 
Recycler →  **Recycle**

The demo itself should tell this story in 2–3 minutes.

The exact demo I would build

Imagine the judge puts an old smartphone on the table.

Scene 1 — The problem

Start with a very short visual:

“What happens to this phone after you throw it away?”

Show:

Household → Informal disposal → Acid/open burning → Environmental & health risk

Then:

“What if we could turn this waste into a traceable recycling transaction?”

Now open your application.

STEP 1 — User scans the e-waste

Screen: **Scan Your E-Waste**

The user clicks:

 **Scan E-Waste**

Take/upload a photo of an old smartphone.

Your AI processes it.

Result:

Detected: Smartphone
Confidence: 94.7%

Then show:

Device Category: Mobile Electronics

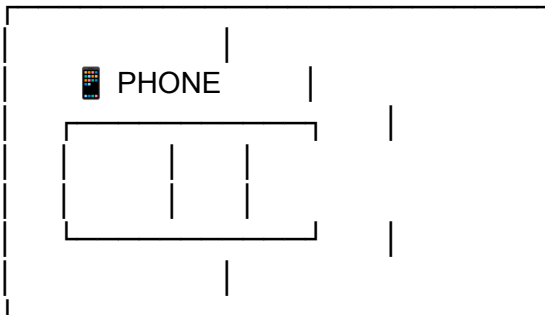
Condition: Used / Damaged

Estimated Age: ~4 years

Visual effect

Don't just display text.

Put a bounding box around the phone:



AI DETECTED

Smartphone

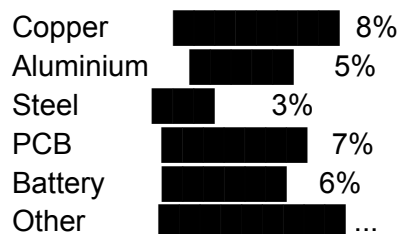
94.7% confidence

This is your first **wow moment**.

● STEP 2 — AI breaks the device down

Now the system says:

♻️ Potential Recoverable Materials



And show a simplified visual of the phone:

Smartphone

↓

PCB

Battery

Copper

Aluminium

Other materials

Don't claim that your computer vision has physically identified these materials from a normal photograph. Instead, the AI identifies the **device type**, and your **device/material knowledge base** estimates likely recoverable materials.

That distinction will protect you from a judge's technical attack.

STEP 3 — VALUE ESTIMATION

Now give the user something genuinely useful.

Estimated Recycling Value

₹420 – ₹650

Breakdown:

Component	Estimated value
PCB	₹280
Aluminium	₹45
Copper	₹70
Other recoverables	₹25
Estimated total	₹420–₹650

Then show:

 **Final value depends on actual weight, condition and recycler rates.**

This makes your system more credible.

STEP 4 — Environmental Impact

Now make the result emotionally meaningful.

 **If recycled responsibly:**

Estimated material recovery: 160–220 g

Potential landfill diversion: 0.4 kg

Hazardous disposal avoided: Battery + electronic components

Then:

RECYCLING IMPACT

Instead of throwing it away → recover its materials.

Don't invent highly precise CO₂ numbers unless you have a verified methodology. For the prototype, label these as **estimates**.

STEP 5 — CREATE THE DIGITAL E-WASTE PASSPORT

This should be one of your signature features.

The system generates:

E-WASTE PASSPORT

EW-2026-000184

Device
Smartphone

Category
Small Electronics

Estimated Value

₹420–₹650

Materials

PCB • Copper • Aluminium • Battery

Status

READY FOR COLLECTION

And generate a **QR code**.

This QR is the digital identity of the recycling transaction.

STEP 6 — Doorstep Pickup

User clicks:

 **Schedule Pickup**

Show:

Pickup Address

[User Location]

Preferred Time

[Tomorrow | 10 AM – 12 PM]

Estimated Value

₹420 – ₹650

Recycler

GreenCycle Recycler

[SCHEDULE PICKUP]

Click.



Pickup Scheduled

Pickup ID: PU-10482

Recycler: GreenCycle
Time: Tomorrow, 10–12

STEP 7 — SWITCH TO THE RECYCLER DASHBOARD

This is where your demo becomes significantly stronger.

Don't remain in the customer application.

Switch to:

Recycler Dashboard

The recycler receives:

 **New Pickup Request**

Pickup ID: PU-10482

Device:
Smartphone

Estimated Value:
₹420–₹650

Location:
2.4 km away

Status:
Awaiting Pickup

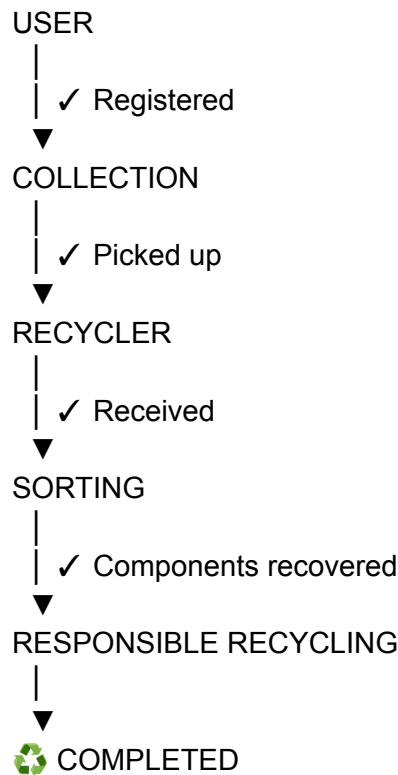
[ACCEPT]

Recycler clicks:

ACCEPT

STEP 8 — Track the recycling chain

Now display:



And this is where your **Digital E-Waste Passport** becomes useful.

The QR code remains associated with the entire lifecycle.

STEP 9 — The killer demo moment

Now deliberately create a **problem scenario**.

Suppose the recycler doesn't update the status.

Your system says:

 **RECYCLING CHAIN INCOMPLETE**

Then:

E-Waste ID: EW-2026-000184

✓ Registered

- ✓ Collected
- ✓ Recycler Received
- ✗ Recycling Confirmation Missing

ACTION REQUIRED

This shows that your platform isn't simply:

“AI identifies a phone.”

It's actually creating a **traceable e-waste workflow**.

STEP 10 — Final result

Complete the recycling step.

Dashboard changes to:

RESPONSIBLY RECYCLED

E-Waste ID
EW-2026-000184

Device
Smartphone

Recycler
GreenCycle

Status
✓ Recycled

Materials Recovered
PCB
Copper
Aluminium

Certificate
[[VIEW DIGITAL CERTIFICATE](#)]

And the user gets:

Recycling Certificate

Your electronic device has been successfully processed through the registered recycling workflow.



Your complete 2–3 minute demo

I would structure the actual video/presentation like this:

0:00–0:15 — Problem

Old phone → improper disposal → hazardous informal processing.

0:15–0:30 — Scan

 Upload phone image.

AI → Smartphone detected

0:30–0:50 — Intelligence

Materials + approximate value + environmental impact.

0:50–1:10 — E-Waste Passport

Generate unique digital identity + QR.

1:10–1:30 — Pickup

User schedules doorstep collection.

1:30–1:50 — Recycler

Recycler accepts request.

1:50–2:10 — Traceability

Collection → Recycler → Sorting → Recycling.

2:10–2:25 — Chain-break demonstration

Show what happens if a stage is missing.

2:25–2:40 — Completion

 Recycling confirmed.

2:40–3:00 — USP

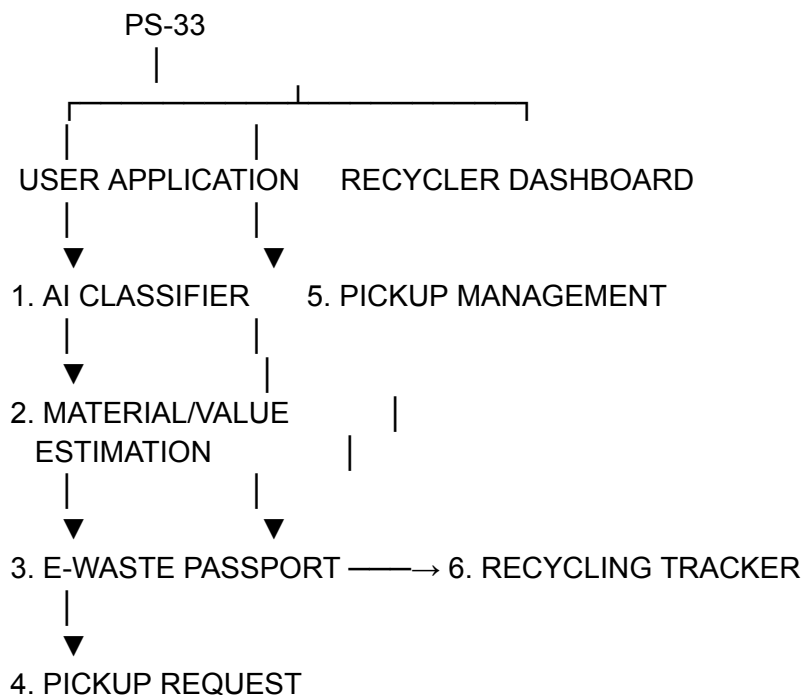
Finish with:

“We don’t just tell people what their e-waste is. We turn it into a valued, traceable recycling transaction.”

Now let's decode what you actually need to BUILD

Don't try to build everything simultaneously.

Build it as **6 modules**:



Module 1 — AI Classifier

Start with **5–10 e-waste categories**, not every possible device.

For example:

- Smartphone

- Laptop
- Monitor
- Keyboard
- Printer
- Battery
- Earphones
- Charger

Input: photograph

Output: category + confidence.

Module 2 — Material + Value Engine

Don't make AI predict exact material quantities.

Instead:

AI:

Smartphone



Knowledge Base:

Typical smartphone material composition



Estimated:

Material recovery + value range

This is technically much more defensible.

Module 3 — Digital E-Waste Passport

Generate:

Unique E-Waste ID

Device

Category

Estimated value

Estimated materials

User

Date

Status
QR

This is the backbone of your workflow.

Module 4 — Pickup

Build:

User
↓
Pickup request
↓
Location
↓
Time slot
↓
Recycler assignment

For the hackathon, use **mock/simulated recycler data**.

Module 5 — Recycler Dashboard

Recycler sees:

NEW
↓
ACCEPTED
↓
PICKED UP
↓
RECEIVED
↓
SORTING
↓
RECYCLED

This makes the demo feel like an actual ecosystem rather than an AI experiment.

Module 6 — Traceability + Analytics

Admin dashboard:

Total E-Waste Registered	1,284 kg
Collected	842 kg
Successfully Recycled	719 kg
Pending	83 kg
Chain Breaks	11

Then:

Top E-Waste Categories

 Smartphones — 38%

 Laptops — 24%

 Monitors — 17%

 Batteries — 12%

Other — 9%

This is where you demonstrate **system-level impact**.

One thing I would change from your original concept

Don't tell judges:

“AI identifies the recyclable components from a photograph.”

That's technically questionable.

Instead say:

“Our AI identifies the e-waste category from an image, and our valuation engine combines that classification with device-specific material profiles and current recycler pricing data to estimate recoverable value.”

Much more technically credible.



And this becomes your real USP

Your project has **three layers**:

Layer 1 — AI

What is this e-waste?



Layer 2 — Intelligence

What could it be worth and what can potentially be recovered?



Layer 3 — Action

How can I get it responsibly recycled and track the process?

That gives you:

AI → VALUATION → ACTION → TRACEABILITY

And **that's the demo I would build for SIH.**

If you want to actually build this in **10–14 days**, the next logical step is to break these six modules into a **Day 1 → Day 14 development plan**, including exactly what your frontend, backend, AI, database and demo data need to contain.

Yes — I understand now. You are **not asking me to design a demo inspired by the video**. You want to **actually build the web app shown in this exact video**, with the same user journey and screens.

I inspected the uploaded video. It is a ~10-second concept/demo showing this flow:

Physical smartphone → camera scan → AI identification → recoverable materials → estimated value + CO₂ saved → doorstep pickup → pickup confirmation/map → Recycler Dashboard.

So from now on, we should treat **this video as the UI/product reference**.

The app we need to reproduce

1. 📱 User Home / Scan

The user points the phone camera at an e-waste item.

We need:

- Camera/scan interface
- Scanning animation
- Device detection
- AI confidence indicator
- Detected device name

2. 🔍 Device Identification Result

The video shows something like:

Device Identification

Smartphone

Then:

Recoverable Materials

- Metals
- PCBs
- Battery
- potentially more

And an **Estimated Value**.

3. ♻️ Environmental Impact

The video prominently shows:

₹2.3kg CO₂ saved

This becomes an important visual part of the product.

4. 💰 Valuation

Example shown in the video:

₹450

with an additional comparison/estimated value.

5. 🚚 Doorstep Pickup

A prominent CTA:

Schedule Doorstep Pickup

Then the user gets a pickup confirmation.

6. 📍 Pickup Confirmation

The video shows:

Pickup Confirmed for Tomorrow, 10 AM

with a map/location component and pickup details.

7. 🏠 Recycler Dashboard

This is the second major side of the application.

The desktop dashboard contains:

- Recycler Dashboard
- Requests
- Search
- Statistics/cards
- Incoming e-waste requests
- Device information
- Value
- Recoverable materials
- Pickup status
- Actions for recycler

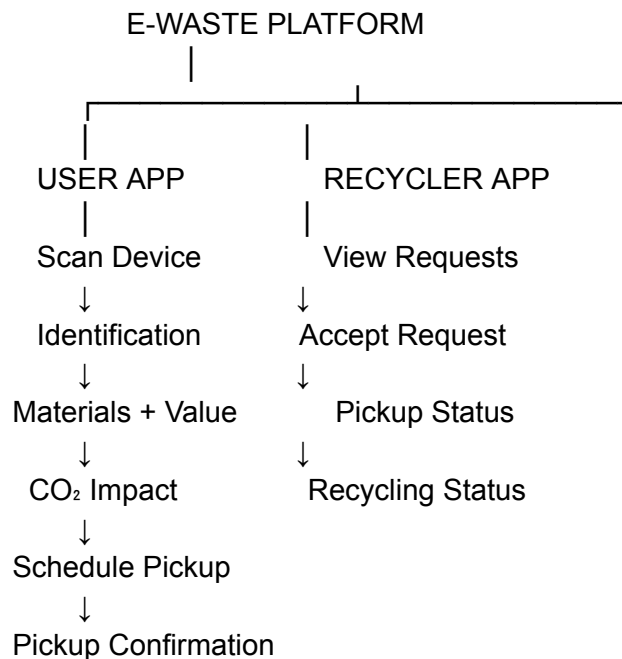
And prominently:

ID#4932 – Smartphone – Ready for Pickup – ₹450

The key point

We should **not overbuild the first version**.

The video is essentially demonstrating **two connected applications**:



And importantly:

We should build the **same visual experience first**, then connect the functionality behind it.

I can help you build this **screen-by-screen from the video**, including the exact UI structure, pages, buttons, cards, navigation, data flow, mock AI response, QR/e-waste passport, pickup flow and recycler dashboard.

Next step should be the UI breakdown: I'll map the video into the exact pages/components we need to create, and then we can start building the web app one screen at a time.